

Xiaoyong (William) Ni
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EDUCATION

University of Glasgow, Glasgow, UK

University of Electronic Science and Technology of China (UESTC), Chengdu, China

Dual Degree program in Electronics and Electrical Engineering

Sept. 2016 – July. 2020

BEng in Communication Engineering

GPA: 3.96/4.0 (Rank: 1/126)

ETH Zurich, Switzerland

Sept. 2020 – Jan. 2024

Master of Science in Electrical Engineering and Information Technology

Research Assistant for 1.5 Years

GPA: 5.5/6.0

EPFL, Switzerland

June. 2024 – Present

PhD in NeuroX institute, brain signal processing and decoding

PUBLICATIONS

Physics Constraints and Expert Regularization for Unsupervised Stimulation Artifact Removal, EMBC 2026, **oral**.

iNeuroTune: Interpretable Neural Data Selection for Unsupervised Decoder Fine-Tuning, EMBC 2026, **oral**.

Exploring the trade-off between deep-learning and explainable models for brain-machine interfaces, NeurIPS 2024.

Adaptive KalmanNet: Data-Driven Kalman Filter with Fast Adaptation, ICASSP 2024.

RTSNet: Learning to Smooth in Partially Known State-Space Models, IEEE TSP 2023.

RTSNet: Deep Learning Aided Kalman Smoothing, ICASSP 2022.

KalmanNet: Neural network aided Kalman filtering for partially known dynamics, IEEE TSP 2022, **IEEE Signal Processing Best Paper Award Nomination**.

Unsupervised Learned Kalman Filtering, IEEE EUSIPCO 2022.

Age of Information for Wireless Energy Harvesting Secondary Users in Cognitive Radio Networks, IEEE MASS 2019.

RESEARCH EXPERIENCE

Integrated Neurotechnologies Lab (INL), EPFL

06/2024- Present

PhD Candidate

- Designing a denoising algorithm to cancel the stimulation artifacts in the recorded brain signals during stimulation treatments
- Designing decoding algorithms to decode from intracranial neural signals to movements, with the application to Brain Machine Interface (BCI) helping paralyzed patients.

Signal and Information Processing Lab (ISI), ETH Zurich

Master Thesis

11/2022- 05/2023

- Researched signal filtering with unknown and time-varying noise distributions.
- Fine-tune (technique borrowed from Large Language Models) the data-driven part of KalmanNet to maintain model-based properties as well as adapting to different context given prior data.

Paid Research Assistant

11/2021- 10/2022 and 09/2023- 01/2024

- Participated in the idea formulation and conducted the simulations for KalmanNet.
- Apply the idea of model-based deep learning to smoothing problems and proposed RTSNet.
- Extended KalmanNet to be adaptive to fast varying noise distributions.

Wireless Communication and Networking Lab (WCAN), Penn State University

07/2019- 10/2019

Research Assistant

- Derived the expression of Age of Information as a function of the packet arrival rate, transmission probability and outage probability in a energy harvesting cognitive radio networks.
- Built a Markov Decision Process model for heterogeneous traffic network, simulated it using Relative Value Iteration method, and proved the optimality of greedy policy in a special case.

Data and Energy Integrated communication Network Lab (DEIN), UESTC

02/2019-06/2019

Research Assistant

- Derived the asymmetric PSK modulation design with nonlinear energy harvester model to enlarge Simultaneous Wireless Information and Power Transfer (SWIPT) rate-energy region.
- Extended the asymmetric modulation design for QAM using the nonlinear energy harvester model and derived the expressions for asymmetric QAM in terms of information and powertransfer.

HONORS AND AWARDS

05/2026	ICML gold reviewer
05/2023	Master's Thesis: 5.75/6.0 (Excellent)
03/2022	IEEE Signal Processing Society Best Paper Award Nomination
06/2020	Outstanding Graduates of Sichuan Province
10/2019	Department's Outstanding Academic Scholarship 2018-2019 (Top 1%)
10/2018	Department's Outstanding Academic Scholarship 2017-2018 (Top 1%)
10/2017	Department's Outstanding Academic Scholarship 2016-2017 (Top 1%)
06/2018	First Prize of UESTC's "Ecovacs" Robot Competition (1/143)
05/2018	Outstanding Member of UESTC's Student Association

SKILLS

Software: Python, MATLAB and C Language

Languages: Chinese (Native), English (Fluent)

Hobbies: Basketball, Swimming, Skiing, Badminton, Table Tennis